



Report

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Summary

① Tasks Status

② Overview of Learned Concepts

③ Selected Paper

Tasks Status

Task	Status	Link
Analisar modelo		Notebook
↪ Resíduos	✓	
↪ Estudo das incertezas correlacionadas e não correlacionadas, experimentais e preditas	🌙	

Tasks Status

Task	Status	Link
Escrever projeto		Projeto
↪ Definir escopo e objetivos	✓	
↪ Esboçar introdução (motivação + observáveis)	✓	
↪ Cronograma	✓	
↪ Selecionar bibliografia	✓	
↪ Listar método/dados (GP, Inferencia Bayesiana)	🌙	
↪ Listar objetivos	○	

Recent studies

Presentation	Status	Link
Exploring the universality of scaled particle spectra with Bayesian inference in heavy-ion collisions	✓	Link
Calibrated (3+1)D hydrodynamic medium for ultrarelativistic collision physics HotJets 2025	✓	Link
Shen - Studying “the most perfect fluid”	✓	Link

Recent studies

Toolbox	Status	Link
2. Gaussian Processes		Lecture Notes

Chapter	Status	Link
QFT for gifted amateur - Chapter 01		Lagrangian
QFT for gifted amateur - Chapter 16		Green Function
QFT for gifted amateur - Chapter 17		Propagators

Selected Paper

Observation of flow vector fluctuations in p–Pb collisions at

$$\sqrt{s_{NN}} = 5.02 \text{ TeV}$$

ALICE Collaboration*

Abstract

Measurements of transverse momentum (p_T) and pseudorapidity (η) dependent flow vector fluctuations in p–Pb collisions at $\sqrt{s_{NN}} = 5.02$ TeV at the CERN Large Hadron Collider are presented. By studying long-range two-particle correlations with a template fit method, potential biases from non-flow effects such as jets and resonance decays are effectively suppressed. Significant p_T - and η -dependent fluctuations of the second-harmonic flow vector are observed with more than 5σ confidence in p–Pb collisions, similar to the observations in Pb–Pb collisions. The influence of residual non-flow effects has been evaluated and cannot account for the observed fluctuations, thereby confirming the observation of flow vector fluctuations in small collision systems at the LHC. Comparisons to model calculations from 3DGlauber+MUSIC+UrQMD and the parton transport model from AMPT are also presented. The measurements provide constraints on the theoretical modelling of the three-dimensional initial geometry and its event-by-event fluctuations, offering critical insights into the origin of collective flow in small collision systems at the LHC.

Figure: Studying long-range two-particle correlations with a template fit method.